Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/29/2005
PI Name:	Meystre, Pierre Ph.D.		
Project Title:	Atom optics in controlled and microgravity environments		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	pierre.meystre@optics.arizona.edu	**Fax:**	FY 520-621-4358
PI Organization Type:	UNIVERSITY	**Phone:**	520-621-4651
Organization Name:	University of Arizona		
PI Address 1:	Optical Sciences Center		
PI Address 2:			
PI Web Page:	http://www.optics.arizona.edu/		
City:	Tucson	**State:**	AZ
Zip Code:	85721	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	04/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2921		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity		

Task Description:	The goal of this project is to carry out theoretical investigations of atom optics in environments characterized by controlled and adjustable accelerations and in field-free situations. The main thrusts of the research are: the theoretical study of nonlinear atom optics, and in particular matter-wave mixing phenomena, in accelerated frames; the investigation of the evolution of matter-wave coherence in accelerated atomic de Broglie waves, with applications in rotational and gravitational sensors; the detailed theoretical analysis of atom holography in microgravity environments, including the use of accelerated beams for the reconstruction process; and a broad theoretical investigation of field-sensitive situations that cannot realistically be performed in earhtbound laboratories. Atom optics is on the verge of becoming an engineering science with broad potential applications, in particular in rotation sensors, gravimeters and gravity gradiometers. Other possible applications include nanofabrication, atom holography, etc. In many cases, the atomic source will be a quantum degenerate gas, such as a Bose condensate --- the atom optical analog of a laser --- or, in the future, a Fermi-degenerate atomic system. Practical devices require however matter waves of short wavelengths, which can be realized by particle acceleration. In addition, external fields used e.g. to compensate earth gravity can lead to random frequency shifts and decoherence that must be avoided in precision measurements. For these and other reasons, it is important to study atom optics in controlled and microgravity environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In the past months, we have made significant progress on several fronts, as evidenced in particular by our list of publications. In the following we concentrate on two highlights of our recent work. In a continuation of our previous research, we have now gained a detailed understanding of the dynamics of Bose-Einstein condensates in double-well potentials, in particular in situations when such potentials are tilted by a gravitational field. We have shown how the interference pattern resulting from the subsequent free fall of the condensate once the double-well potential is turned off can be exploited to measure small forces, in particular the acceleration of gravity. We now have a good understanding of ways to exploit the quantum phase-like transitions that appear in these systems as the ratio between mean-field energy and tunneling is varied cto improve the sensitivity of these devices. This work is continuing, with the goal of developing further "sub shot-noise" detection schemes. In particular, it will be extended to fermionic atomic systems: we have previously shown that when compaqred to bosons, fermionic atoms can be of significant potential advantage in the detection of very weak forces, in a situation reminiscent to the use of white-light interferometry to measure small phase shifts in conventional optics. The potential advantage of fermions in sepcific applications has also been demonstrated in experimental work by the group of M. Inguscio in Italy. Maybe the most important new development in our research, though, is the study of molecular optics. This is a natural extension following the experimental realization of molecular condensates by serval groups in the last yar or so. While much of the more fundamental in this direction is currently directed toward understanding of long-standing problems such as e.g. the BEC-BCS cross-over, ultracold molecules also offer much potential for applications, in particular in sensor development and quantum information technology. Our own work here is proceeding along several fronts: first, we are developing an undertsanding of the dynamics of molecule formation, both in situations involving Feshbach resonances and photoassociation. In addition, we are also studying molecule detection and counting techniques to understand the quantum statistical properties of the molecular fields produced from bosonic atoms, fermionic atoms, and mixtures of bosonic and fermionic atoms, and eventually to develop novel detection schemes. In this context we are particularly interested in understanding the formation, manipulation, control and detection of ultracold heteronuclear molecules. There is now little doubt that substantial samples of such molecules in their ground electro-vibrational state can be produced, for instance by photoassociation, and these molecules should then be easy to manipulate further owing to their permatent electric dipole moment. Based on experiments by the groiup of G. Meijer in Berlin, we expect that it should be possible to accelerate these molecules (which would remain cold in their moving frame) and inject them in molecular storage rings of centimeters in diameters. This offers fascinating prospects for matter-wave rotation sensors order-of-magnitude more sensitve than is presently the case. However, this will only be possible if decoherence mechanisms can be understood and kept under control. It will also be necessary to develop molecule detectors at the single-molecule level to take full advantage of the extraordinary potential of these molecules. Work in these directions is now a significant component of our efforts, and significant progess is expected in the coming year.
Bibliography Type:	Description: (Last Updated: 02/24/2006)
Articles in Peer-reviewed Journals	Meiser, D; Search, C. P.; Meystre, P. "Diffraction of ultra-cold fermions by a quantized light field: standing versus traveling waves" Phys. Rev. A 71, 013404 , Jan-2005
Articles in Peer-reviewed Journals	Meystre, P. "Nonlinear matter-wve optics of bosons and fermions" J. Phys. B 38, S617 , May-2005
Articles in Peer-reviewed Journals	Meystre, P. "From micromasers to aotm optics and back" Laser Physics 15, 99 , Jan-2005
Articles in Peer-reviewed Journals	Miyakawa, T.; Meystre, P. "Boson-fermion coherence in a spherically symmetric harmonic trap" Phys. Rev. A 71, 033624 , Mar-2005
Articles in Peer-reviewed Journals	Meiser, D.; Meystre, P. "Number statistics of molecules formed by ultracold atoms" Phys. Rev. Lett. 94, 093001 , Mar-2005
Articles in Peer-reviewed Journals	Jaaskelainen, M.; Meystre, P. "Transient dynamics of Bose-Einstein condensates in double-well potentials" Phys. Rev. A 71, 043603 , Apr-2005
Articles in Peer-reviewed Journals	Meiser, D; Search, C. P.; Meystre, P. "Molecules formation as a diagnostic tool for second-order correlations of ultracold gases" Phys. Rev. A 71, 033621 , Mar-2005
Articles in Peer-reviewed Journals	Dutta, O.; Jaaskelainen, M.; Meystre, P. "Polarizing beam splitter for dipolar molecules" Phys. Rev. A 71, 051601 (R) , May-2005

Presentation	Meystre, P. "When atoms become waves" Distinguished Lecture Series, CREOL, College of Optics, University of Central Florida Feb-2005
Presentation	Meystre, P. "Quantum optics of molecules" Invited paper, International Symposium ofn “Frontiers in Quantum Optics and Applications,“, Munich, Germany Apr-2005
Presentation	Meystre, P. "Quantum optics of ultracold molecular dimers" Invited paper, DAMOP Annual Meeting, Lincoln, NE May-2005
Presentation	Uys, H.; Miyakawa, T.; Meiser, D.; Meystre, P. "Passage time statistics in the formation of ultracold dimers from fermionic atoms" Contributed paper, DAMOP Annual Meeting, Lincoln, NE May-2005
Presentation	Miyakawa, T.; Meystre, P. "Dynamics of association in ultra-cold Fermi gases in spherically symmetric harmonic traps" Contributed paper, DAMOP Annual meeting, Linclon, NE May-2005
Presentation	Dutta, O.; Jaaskelainen, M.; Meystre, P. "A polarizing beam splitter for dipolar molecules" Contributed paper, DAMOP Annual Meeting, Linclon, NE May-2005
Presentation	Jaaskelainen, M.; Meystre, P. "Dynamics of Bose-Einstein condensates in double wells" Contributed paper, DAMOP Annual Meeting, Lincoln, NE May-2005